Shell–and–Tube Heat Exchangers For General Refinery Services

API STANDARD 660
FIFTH EDITION, JUNE 1993

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Shell–and–Tube Heat Exchangers For General Refinery Services

Manufacturing, Distribution and Marketing Department

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FOREWORD

This standard is based on the accumulated knowledge and experience of petroleum refiners, shell-and-tube heat exchanger manufacturers, and engineering contractors. The objective of this standard is to provide a purchase specification to facilitate the procurement and manufacture of shell-and-tube heat exchangers for general refinery services.

This standard requires the purchaser to specify certain details and features. Although it is recognized that the purchaser may desire to modify, delete, or amplify sections of this standard, it is strongly recommended that all modifications, deletions, and amplifications be made by supplementing this standard rather than by rewriting or incorporating sections into another complete standard.

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Shell-and-Tube Heat Exchangers for General Refinery Services

SECTION 1—GENERAL

1.1 Scope

1.1.1 This standard defines the minimum requirements for the mechanical design, material selection, fabrication, inspection, testing, and preparation for shipment of shell-and-tube heat exchangers for general refinery services.

1.1.2 As used in this standard, the term *heat exchangers* includes condensers, coolers, and reboilers.

1.2 General

Note: A bullet (●) in the margin indicates that a decision by the purchaser is required. Decisions should be indicated on the Shell-and-Tube Heat Exchanger Checklist (see Appendix B)

1.2.1 Exchangers shall conform to the latest edition of the *Standards of the Tubular Exchanger Manufacturers Association*[1] (TEMA Standards) and to the additional requirements of this standard. Compliance with the latest edition of Section VIII, Division 1, of the ASME[2] *Boiler and Pressure Vessel Code* is mandatory.

1.2.2 The vendor shall comply with local rules and regulations that are specified by the purchaser.

● **1.2.3** If specified by the purchaser or required by state or local codes, the vendor shall register each exchanger with the National Board of Boiler and Pressure Vessel Inspectors.

1.3 Referenced Publications

Unless otherwise specified, the most recent editions or revisions of the following standards or codes shall, to the extent specified herein, form a part of this standard:

ASME
- B1.20.1 *Pipe Threads, General Purpose*
- B16.5 *Pipe Flanges and Flanged Fittings*
- B16.11 *Forged Steel Fittings, Socket-Welding and Threaded*
- B16.47 *Large Diameter Steel Flanges: NPS 26 Through NPS 60*
- *Boiler and Pressure Vessel Code*, Section II, "Material Specifications"; Section V, "Nondestructive Examination"; Section VIII, Division 1, "Rules for Construction of Pressure Vessels"; Section IX, "Welding and Brazing Qualifications"

AWS[3]
- D1.1 *Structural Steel Welding Code*

TEMA
- *Standards of the Tubular Exchanger Manufacturers Association*

1.4 Definition of Terms

Terms used in this standard are defined as follows:

a. An *exchanger unit* consists of one or more exchangers for a specified service that may include alternative operating conditions.
b. The *item number* is the purchaser's identification number for a unit.
c. The *effective surface* is the outside surface area of the tubes, excluding the portion in each tubesheet.

1.5 Conflicting Requirements

1.5.1 In case of conflict between documents, the order of priority shall be:

a. The inquiry or purchase order.
b. Shell-and-Tube Heat Exchanger Specification Sheet (see Appendix D).
c. The purchaser's standard on heat exchangers.
d. Referenced purchaser standards.
e. API Standard 660.
f. TEMA Standards.
g. ASME *Boiler and Pressure Vessel Code.*

1.5.2 The vendor shall obtain written approval from the purchaser before proceeding with work affected by a conflict between the proposal and the purchase order, unless an exception is stated as such in the purchase order.

[1]Tubular Exchanger Manufacturers Association, 25 North Broadway, Tarrytown, New York 10591.
[2]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.
[3]American Welding Society, P.O. Box 351040, Miami, Florida 33135.

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SECTION 2—PROPOSALS

2.1 Purchaser's Responsibilities

The purchaser's inquiry shall include the specification sheet or sheets, a checklist, and other applicable information outlined in this standard (see Appendices A, B, C, and D). All necessary data for the design of an exchanger unit, including any special requirements or exceptions to this standard, shall be provided.

2.2 Vendor's Responsibilities

2.2.1 The vendor's proposal shall include, for each unit, a completed specification sheet (see Appendices A and D) or, when a specification sheet is included in the inquiry, a statement indicating complete compliance with that specification sheet.

2.2.2 Designs that are not fully defined by the nomenclature in Section 1 of the TEMA Standards shall be accompanied by sketches that are sufficient to describe the details of construction.

2.2.3 Where distributor belts or expansion joints are provided, the vendor shall define the type of construction proposed.

2.2.4 The proposal shall include a detailed description of all exceptions to the requirements of the purchaser's inquiry.

2.2.5 The shipping date shall be determined on the basis of a specified number of weeks after receipt of verbal or written notification of the purchase order. This date should allow sufficient time for transmittal, review, and return of drawings.

2.2.6 The proposal shall state the number of weeks the vendor will require to submit welding procedures.

2.2.7 Unless otherwise specified, one complete set of spare gaskets shall be supplied for each exchanger.

SECTION 3—DRAWINGS AND OTHER REQUIRED DATA

3.1 Outline Drawings

● **3.1.1** Within 4 weeks after receipt of verbal or written notification of the purchase order, the vendor shall submit for review the number of copies of outline drawings specified by the purchaser for each exchanger unit. The drawings shall include the following information:

a. The service, purchaser's item number, project name and location, purchaser's order number, vendor's shop order number, and other special identification numbers.
b. Design pressure, test pressure, design temperature, minimum design metal temperature, and any restriction on testing or operation of the exchanger.
c. Maximum allowable working pressure (MAWP) in the corroded conditions and at design temperature for the shell side and tube side.
d. Connection sizes, location, orientation, projection, direction of flow, and, if flanged, the rating and facing.
e. Coupling sizes, rating, and orientation.
f. Dimensions, orientation and location of supports, including bolt holes and slots, and the stacking arrangement.
g. Overall dimensions of the exchanger.
h. Tube-bundle removal clearance.
i. Weights of the exchanger, empty and full of water, and of the tube bundle.
j. Specified corrosion allowance for each side of the exchanger.
k. Reference to the applicable code and the purchaser's purchase specification.
l. Requirements for postweld heat treatment.
m. Requirements for radiographic examination.
n. Requirements for material impact testing.
o. Surface preparation and painting.
p. Gasket material.
q. Insulation thickness on the shell side and tube side.
r. Location of expansion joints, vapor distributors, and any other specialty components or closures.

3.1.2 Review of outline drawings by the purchaser shall not relieve the vendor of his or her responsibility to meet the requirements of the purchase order.

3.2 Information Required After Drawings Are Reviewed

3.2.1 Following the purchaser's review of the outline drawings, the vendor shall furnish one reproducible copy of the certified outline drawings.

3.2.2 Gasket details (including type and material) shall be shown on a separate drawing. This drawing shall not be marked with any restrictions for use.

3.2.3 Qualified welding procedures and procedure qualification records, as required by Section IX of the ASME Code, shall be submitted for review.

3.2.4 Upon receipt of the purchaser's review comments on the outline drawings, the vendor shall submit copies of all detail drawings for the purchaser's review.

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The detail drawings shall fully describe the exchanger and shall include, but not be limited to, the following information:

a. Full and cross section views with all dimensions and materials sufficient for stress calculations for each part.
b. Bundle details, including tube layout; tube description and number in each pass; cross baffle cut, layout, and orientation in a view that shows the cuts; details and locations of all bypass seals; and tubesheet and tube-hole details, including cladding when required.
c. Details of each pressure retaining weld, including weld material, weld location, and applicable nondestructive examination method.
d. Details of each weld for nonpressure attachments.
e. Complete bills of materials, including the ASME specification number.
f. Expansion joint details.
g. Details of cladding and weld overlay.
h. One reproducible copy of a weld map for each exchanger showing the weld joints, welding procedure numbers, and the nominal thickness and ASME specification number of each component.
i. A single reproducible drawing for description of welded tube-to-tubesheet joints referencing a procedure for installation, welding, expansion, inspection, and testing.
j. Flange-face finish.

3.2.5 The vendor shall submit for the purchaser's review the following documentation:

a. Two copies of all mechanical design calculations for all the exchanger pressure-retaining components. Where calculations are made on a computer, all input and output data shall be detailed so as to facilitate an understanding of the calculation procedures; also, the formulas in the applicable sections of the ASME Code, Section VIII, Division 1, and the TEMA Standards shall be referenced.
b. Design calculations based on seismic, transportation, and/or piping loads, when these loads are provided by the purchaser.
c. Design calculations for supports or lifting and pulling devices, when requested by the purchaser.
d. Vibration analysis, when specified by the purchaser.
e. One reproducible copy of the proposed procedures for assembly of flange joints, where controlled bolt tightening procedures (such as hydraulic torque wrenches) are used. Identify any lubricants used.

3.2.6 Upon final review, vendor shall revise all the drawing and welding procedures as required and submit two copies of each with the following marked on every sheet separately and dated: "CERTIFIED FOR CONSTRUCTION."

3.3 Reports and Records

Within 10 days after the exchanger is completed, the vendor shall furnish the purchaser with the specified number of copies of the following documents:

a. An "as-built" specification sheet.
b. All outline and detail drawings, marked "CERTIFIED AS-BUILT."
c. A certified record of all impact tests performed.
d. Certified mill test reports for all pressure parts, including tubes (each material test report shall be identified by part number).
e. A complete certified bill of materials suitable for obtaining all replacement parts, including quantity, description, material specification, and identification of each part.
f. Temperature charts of all postweld heat treatments.
g. A completed manufacturer's data report in accordance with Section VIII, Division 1, of the ASME Code.
h. Nameplate rubbings or a facsimile.
i. All mechanical design calculations, marked "CERTIFIED AS-BUILT."
j. One reproducible copy of a nondestructive testing map or procedure for each exchanger showing the radiographic, magnetic-particle, liquid-penetrant, ultrasonic, hardness, impact, and other applicable testing requirements.
k. Tube-joint leak-test results.
l. Hydrostatic-test records in the form of a chart or certification.

SECTION 4—DESIGN

4.1 Design Temperature

4.1.1 All exchangers shall have two design temperatures for each side, a maximum design temperature and a minimum design metal temperature (MDMT), specified by the purchaser in Appendix D.

4.1.2 For external bolting and for components exposed to both shell-side and tube-side fluids, the design temperature shall be the same as the shell-side or tube-side design temperature, whichever is more severe. The design temperature for nonferrous tubesheets shall be subject to purchaser approval.

4.2 Cladding for Corrosion Allowance

Where cladding (including weld overlay) is used, the full thickness of the cladding shall be used as corrosion allowance unless specified otherwise by the purchaser.

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4.3 Shell Supports

4.3.1 The fixed shell support of removable-bundle heat exchangers shall be designed to withstand a longitudinal force equal to 150 percent of the bundle weight applied at the exchanger bundle centerline. The shear stress for supports shall not exceed 40 percent of the yield strength of the material.

4.3.2 Horizontal exchangers shall be provided with two or more saddles designed to support the units under all specified conditions. Design of the saddles shall be as follows:

a. Saddles shall be attached to saddle bearing plates.
b. The bearing provided by the saddles shall extend over at least one-third of the circumference of the shell.
c. Saddle bearing plates shall have the same nominal chemical composition as the shell and shall be continuously welded directly to the exchanger shells.
d. The plates shall be provided with ¼-inch (6-millimeters) diameter vent holes located at the vertical centerline.

4.3.3 The lower shells of stacked removable-bundle heat exchangers shall be designed to carry the superimposed load without suffering distortion that could cause binding of the tube bundles.

4.3.4 The vendor's design shall provide for a shim allowance of approximately ¼ inch (6 millimeters) between the faces of stacked exchanger intermediate supports.

4.3.5 For horizontal exchangers, slotted holes shall be provided in the base plate of all but one of the saddles to allow for longitudinal movement due to thermal expansion or contraction. Slot width shall be equal to anchor bolt diameter plus 5/16 inch (8 millimeters), and slot length shall be equal to anchor bolt diameter plus allowance for longitudinal movement plus 5/16 inch (8 millimeters).

4.4 Stationary Head

4.4.1 When a bonnet (see TEMA Standards, Figure N-1.2, Type B stationary head) is provided, the design of the exchanger shall permit full design pressure hydrotesting on the shell side with the bonnet removed.

4.4.2 Stay bolts shall not be used.

4.5 Floating Head

4.5.1 Floating-head cover bolting shall conform to the requirements of Section 5, Paragraph RCB-11, of the TEMA Standards. Bolt spacings and clearances shall not be less than the TEMA Standards' recommended minimum.

4.5.2 Floating-head cover bolting shall be readily accessible and shall have adequate wrench clearance when the shell cover is removed.

4.5.3 Packed floating-head tailpipe and packed floating tubesheet designs are not acceptable.

4.5.4 Unless otherwise agreed between purchaser and vendor, floating heads shall be designed for full design pressure on either side, with atmospheric pressure or specified vacuum on the other side. Design shall be in accordance with Appendix 1, Figure 1-6 (d), of Section VIII, Division 1, of the ASME Code.

4.5.5 Internal floating-head covers are to have the specified corrosion allowance on all wetted surfaces except gasket seating surfaces. A corrosion allowance shall be included on the back side of the floating-head backing device.

4.6 Tube Bundle

4.6.1 TUBES

4.6.1.1 The minimum outside diameter of the tube shall be ¾ inch (19 millimeters) unless specified otherwise by the purchaser.

4.6.1.2 Unless otherwise specified by the purchaser, the wall thickness shall be as listed in Table 1, or heavier if required by the design conditions.

Table 1—Tube Wall Thickness

Tube Material	Tube Wall Thickness		
		inches	millimeters
Carbon Steel, low-alloy steel, aluminum, and aluminum alloy	14 BWG[a]	0.083[a]	2.1[a]
Copper and copper alloys	16 BWG[a]	0.065[a]	1.7[a]
High-alloy steel and other nonferrous materials	16 BWG[b]	0.065[b]	1.7[b]
Titanium	18 BWG[b]	0.049[b]	1.2[b]

Note: BWG = Birmingham wire gage.
[a]Minimum.
[b]Average.

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4.6.1.3 The mean radius of U-bends shall not be less than 1½ times the nominal outside diameter of the tube.

4.6.1.4 The number of tubes in any pass shall not be greater than 10 percent above or below the average number of tubes per pass.

4.6.2 TUBESHEETS

4.6.2.1 The stationary tubesheet on a vertical unit with the channel down shall be provided with a suitable means of holding the bundle in place when the channel is removed. When collar bolts or drilled and tapped holes are used, at least four shall be provided and their location shall be identified.

4.6.2.2 The distance between the edge of the tube holes and the edge of all gasket grooves shall not be less than 1/16 inch (1.5 millimeters) for tubesheets with expanded tube-to-tubesheet joints and not less than ⅛ inch (3 millimeters) for tubesheets with seal- or strength-welded tube-to-tubesheet joints. The tolerance shall be −0.02 inch (−0.508 millimeter).

4.6.2.3 Unless otherwise agreed between purchaser and vendor, tubesheets shall be designed for full design pressure on either side, with atmospheric pressure or specified vacuum on the other side.

4.6.2.4 When a full-diameter tubesheet is required, it shall be designed to meet the requirements of Paragraph UG-34, Section VIII, Division 1, of the ASME Code.

4.6.3 TRANSVERSE BAFFLES AND SUPPORT PLATES

4.6.3.1 The thickness of transverse baffles and support plates shall not be less than the specified shell-side corrosion allowance.

4.6.3.2 Transverse baffles and support plates shall have notches that are ⅜ inch (10 millimeters) high to facilitate drainage.

4.6.4 IMPINGEMENT PROTECTION

4.6.4.1 If necessary, impingement protection shall be provided by a plate baffle on the tube bundle, a distributor belt, or another means agreed upon by the purchaser and the vendor.

4.6.4.2 When used, an impingement plate baffle shall extend at least 1 inch (25 millimeters) beyond the projection of the nozzle bore.

4.6.4.3 The unrestricted entrance area at the impingement plate as well as the shell entrance shall not be less than the flow area of the inlet nozzle.

4.6.4.4 The nominal thickness of the impingement plate baffle shall not be less than ¼ inch (6 millimeters).

4.6.4.5 Perforated impingement plates shall not be used.

4.6.5 BYPASS SEALING DEVICES

Note: See Figure 1 for an example of a typical location for a bypass sealing device.

4.6.5.1 Bypass sealing devices (such as seal bars, dummy tubes, or tie-rods) shall be used for nonisothermal service when bypass clearances exceed 5/16 inch (16 millimeters) and shall be located as follows:

a. When the distance between baffle-cut edges is 6 tube pitches or less, a single seal, located approximately halfway between the baffle cuts, shall be provided.
b. When the distance between baffle-cut edges exceeds 6 tube pitches, multiple seals shall be provided. A seal shall be located every 5 to 7 tube pitches between the baffle cuts, with the outermost seals not more than 3 inches (76 millimeters) from each baffle-cut edge.

4.6.5.2 Peripheral bypass seals shall extend from the peripheral edge of the transverse baffle into the tube bundle so that the clearance to the nearest tube does not exceed the nominal clearance between tubes.

4.6.5.3 Internal bypass seals shall be installed so that the clearance to the nearest tube does not exceed the nominal clearance between tubes.

● **4.6.5.4** Bypass sealing devices shall be located to minimize obstruction of mechanical cleaning lanes or be readily removable. When continuous tube lanes must be maintained in the tube bundle to facilitate cleaning, the purchaser shall specify this requirement.

4.6.5.5 The nominal thickness of seal strips shall be the nominal thickness of the transverse baffles or ¼ inch (6 millimeters), whichever is thinner.

4.6.5.6 Bypass seal strips shall be attached to the transverse baffles by continuous welds on one side of each baffle.

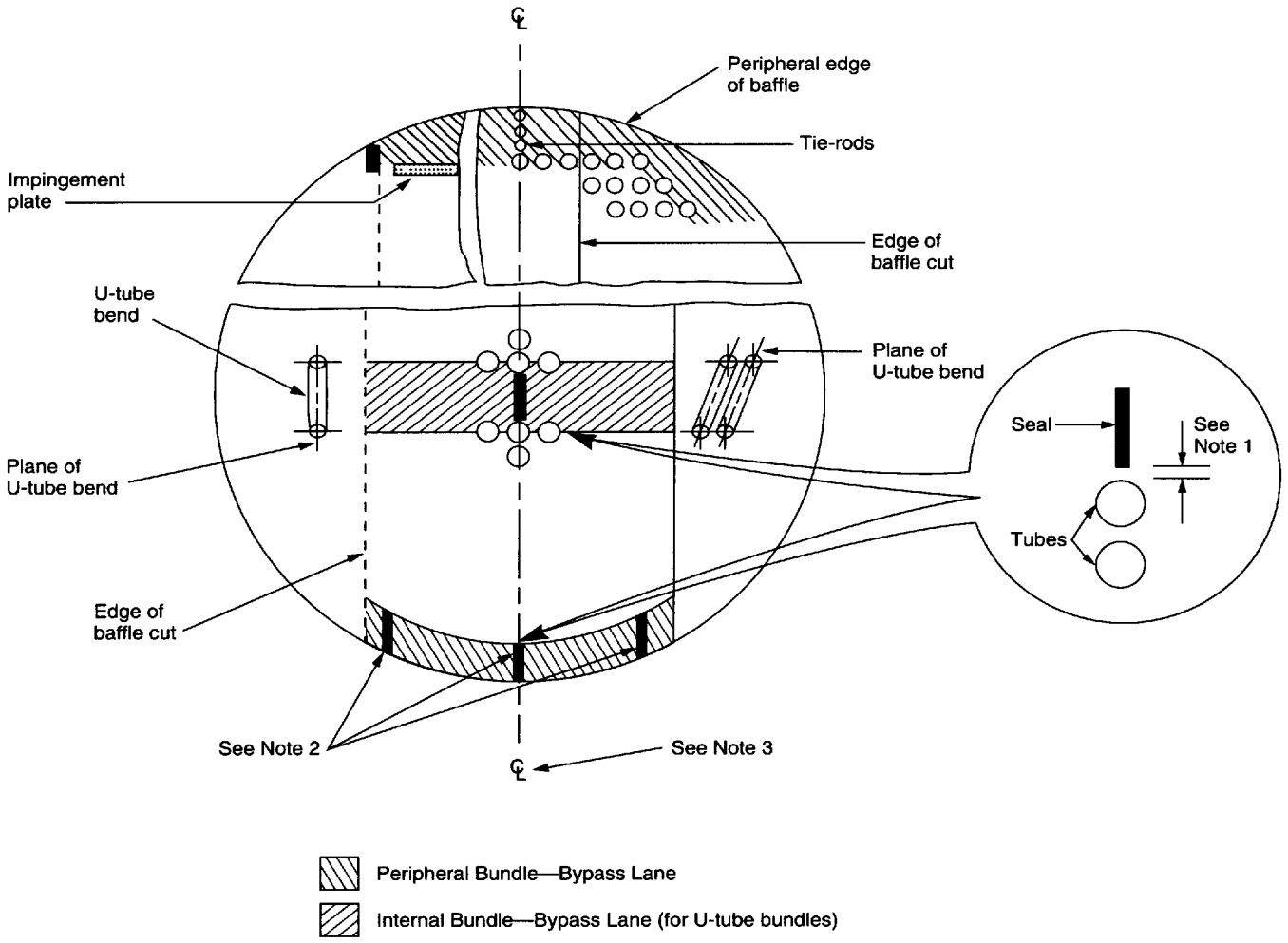
4.6.5.7 Peripheral bypass seal strips shall not restrict the bundle inlet or outlet flows.

4.6.6 BUNDLE SKID BARS

4.6.6.1 For all removable bundles that weigh more than 20,000 pounds (8720 kilograms), a continuous sliding surface shall be provided to facilitate bundle removal. If skid bars are used, they shall be welded to the transverse baffles and support plates to form a continuous sliding surface.

4.6.6.2 Skid bars shall protrude 1/32 inch (0.8 millimeter) beyond the outside diameter of baffle and support plates.

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Notes:
1. Clearance shall not exceed the nominal clearance between tubes.
2. Multiple seals shall be reasonably uniformly spaced.
3. Single seals shall be located on the centerline of the tube bundle.

Figure 1—Typical Cross Sections of Tube Bundle Showing Locations of Bypass Sealing Devices

4.6.6.3 The leading and trailing edges of skid bars and seal strips shall be provided with a radius or a bevel to prevent damage to the shell during bundle insertion and removal.

4.7 Nozzles and Other Connections

4.7.1 Connections NPS 1½ and larger shall be flanged.

● **4.7.2** Connections shall be beveled for welding if specified by the purchaser.

4.7.3 Nonflanged connections smaller than NPS 1½ shall be forged 6000-pound couplings in accordance with ASME B16.11 or integrally reinforced welding outlet fittings with tapered threads in accordance with ASME B1.20.1. Threaded connections shall not be used when hydrogen is present at a partial pressure exceeding 100 pounds per square inch (690 kilopascals), absolute.

4.7.4 Flanged connections shall be of one of the following types:

a. Forged or centrifugally cast long welding neck (integrally flanged).
b. Pipe welded to a forged or centrifugally cast welding-neck flange.
c. Pipe welded to a forged slip-on flange.

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4.7.5 Slip-on flanges shall not be used under any of the following conditions:

a. A design pressure that exceeds 300 pounds per square inch (2070 kilopascals) gauge.
b. A design temperature that exceeds 750°F (400°C).
c. A corrosion allowance greater than ⅛ inch (3 millimeters).
d. When hydrogen is present at a partial pressure exceeding 100 pounds per square inch (690 kilopascals), absolute.
e. Cyclic service.

4.7.6 Steel flanges shall be in accordance with ASME B16.5 or ASME B16.47. Flanges outside the scope of ASME B16.5 and ASME 16.47 shall be designed in conformance with the procedures outlined in Appendix 2 of Section VIII, Division 1, of the ASME Code.

● **4.7.7** The projection of flanged connections shall permit the removal of through-bolting from either side of the flange without removal of the insulation. The insulation thickness shall be specified by the purchaser.

4.7.8 Integrally reinforced nozzles shall be designed so that standard wrenches will fit the nuts without interference from nozzle neck reinforcement.

● **4.7.9** Nozzles shall be designed to allow adequate clearance for use of torquing tools specified by the user.

● **4.7.10** When chemical cleaning connections are specified, the size shall not be less than NPS 2.

4.8 Flanged External Girth Joints

4.8.1 Channel and shell external girth joints shall be of through-bolted construction.

● **4.8.2** Flanges for external girth joints shall be of the forged welding-neck type. Slip-on flanges or other types of forged welding flanges may only be used if approved by the purchaser.

4.8.3 When nubbins are provided on the flange face, they shall be located on the female (grooved) flange.

4.8.4 The nominal clearance between flanges after assembly shall not be less than ⅛ inch (3 millimeters). The clearance between flanges shall extend within the bolt circle to permit checking flanges for radial distortion caused by an excessive bolt load.

4.8.5 Flanges shall be spot or back faced to the extent required by ASME B16.5.

4.8.6 Hardened washers shall be provided under nuts for all bolts having diameters of 1½ inches (38 millimeters) or larger. The washers shall be at least ¼ inch (6 millimeters) thick.

4.9 Gaskets

4.9.1 Gaskets in hydrocarbon service shall be double-jacketed fiber-filled metal, solid metal, or spiral-wound fiber-filled metal.

4.9.2 A means to prevent overcompression of spiral-wound gaskets shall be provided.

4.10 Handling Devices

4.10.1 The lifting device shall be a weld-on lug or ring provided with a hole not less than 1½ inches (38 millimeters) in diameter. Wherever possible, the lug or ring shall be located at the top of the component above its center of gravity; otherwise, two lugs or rings suitably located shall be provided. The lifting device shall be designed to support a load at least twice the weight of the component.

4.10.2 For stacked exchangers, two lifting lugs or rings located at the top, approximately 45 degrees from the vertical centerline, shall be provided on all covers.

4.10.3 Pulling lugs or tapped holes for the insertion of eyebolts shall be provided on the outer face of the stationary tubesheet to aid removal of the bundle from the shell. Pulling lugs and tapped holes shall be designed for a pulling force equal to at least 150 percent of the bundle weight. Each tapped hole shall be fitted with a threaded plug of the same material as the tubesheet face. The exposed section of the plug shall be at least 2 inches (51 millimeters) in length.

● 4.11 Hydrogen Service

Where the tube side and/or shell side will be exposed to hydrogen at a partial pressure exceeding 100 pounds per square inch (690 kilopascals), absolute, totally enclosed spaces between welds shall be eliminated or vented with a ¼-inch (6-millimeters) diameter hole.

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SECTION 5—MATERIALS

5.1 General

In service where stress cracking is possible, internal bolting of ferritic material shall be ASME SA-193, Grade B7M. Such service shall be identified by the purchaser.

5.2 Gaskets

5.2.1 Material for metal-jacketed or solid metal gaskets shall have a corrosion resistance at least equal to that of the gasket contact surface material.

5.2.2 Metal windings of spiral-wound gaskets shall be of austenitic stainless steel material unless specified otherwise by the purchaser.

5.2.3 Solid metal gaskets, including welds, shall be softer than the gasket contact surface.

5.3 Tubes

5.3.1 Integrally finned tubes of copper alloy shall be furnished in the annealed temper (ASME SB-359).

5.3.2 All welded tubes shall receive an eddy current test in the finished condition over the total length of the tubes.

SECTION 6—FABRICATION

6.1 Shells

6.1.1 All longitudinal and circumferential welds of shells for other than kettle-type units shall be finished flush with the inner contour for ease of tube-bundle insertion and withdrawal. For kettle-type units, this requirement is waived for welds in the enlarged section that are not in the bottom quadrant of the shell.

6.1.2 For removable-bundle exchangers, the permissible out-of-roundness of a completed shell, after any required heat treatment, shall allow a metal template to pass through the entire shell length without binding. The template shall consist of two rigid disks (each with a diameter equal to the diameter of the transverse baffle or support plate) rigidly mounted perpendicularly on a shaft and spaced not less than 12 inches (305 millimeters) apart.

6.1.3 Transverse baffle-to-shell clearances greater than indicated in Table RCB-4.3 or RGP-RCB 4.3 of the TEMA Standards shall not be used unless specifically approved by the purchaser.

6.2 Pass Partition Plates

Pass partition plates for forged or welded channels and floating heads shall be welded full length, either from both sides or with full-penetration welds, except for special designs approved by the purchaser. When welded from both sides, the first 2 inches (51 millimeters) from the gasket face shall be full-penetration welds.

6.3 Connection Junctions

Nozzles and couplings shall not protrude beyond the inside surface of the shell, channel, or head to which they are attached.

6.4 Tubes

Each U-tube shall be formed from a single length having no circumferential welds.

6.5 Welding

6.5.1 Welds may be made by any approved welding process (other than oxyacetylene) specified in Section IX of the ASME Code.

6.5.2 Category A and B joints (see Paragraph UW-3 of Section VIII, Division 1, of the ASME Code) shall be as specified in Table UW-12 (No. 1) of Section VIII, Division 1, of the ASME Code.

6.5.3 All welds attaching connections to cylinders or to heads shall fully penetrate the total thickness of the component wall or the connection wall forming the attachment.

6.5.4 Connections abutting a component fabricated from plate (for example, a set-on nozzle) are permitted, provided the edge of the hole in the plate to which the connections are attached shall be examined for lamination by means of a magnetic-particle or dye-penetrant test. Indications found shall be cleared to sound metal and then back welded.

6.5.5 Backing strips that remain in place on the inside of a component after completion of welding, thus forming a crevice, shall not be used unless approved by the purchaser.

6.5.6 For certain process conditions, the purchaser may specify welding of tubes to tubesheets. The welding and testing procedures in these instances shall be mutually agreed upon by the purchaser and the vendor.

6.5.7 Welds attaching insulation support rings need not be continuous.

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6.5.8 Welds attaching other nonpressure attachments (such as lugs or structural steel supports) shall be continuous.

6.5.9 Repair-associated welding procedures shall be submitted to the purchaser for review before the start of repair.

6.5.10 Full-penetration welds are required for those components that will be exposed to hydrogen at a partial pressure exceeding 100 pounds per square inch (690 kilopascals), absolute.

6.6 Heat Treatment

6.6.1 Machined contact surfaces, including any threaded connections, shall be suitably protected to prevent scaling or loss of finish during heat treatment.

● **6.6.2** Requirements and procedures for heat treatment after bending for the bend portion of U-tubes shall be specified by the purchaser.

6.6.2.1 When the purchaser specifies heat treatment of U-bends of austenitic stainless steel, the procedure shall be as described by the ASME Code, Section II-A SA-688, Paragraph 7.2.

6.6.2.2 The U-bends of copper and copper alloy tubes, including copper-nickel alloys, shall be heat treated as required by the ASME Code, Section II-B SB-395, Paragraph 7.5.

6.6.3 The heat-treated portion of the U-bend shall extend at least 6 inches (152 millimeters) beyond the point of tangency.

6.6.4 Postweld heat treatment of fabricated carbon steel channels and bonnets is required for the following:

a. Channels and bonnets with six or more tube passes.
b. Two- or four-pass channels and bonnets whose cylinder-to-nozzle diameter ratios are 2.0 or less.
c. One-pass channels and bonnets whose nozzles are not axial and whose cylinder-to-nozzle diameter ratios are 2.0 or less.

● **6.6.5** Weld-overlaid carbon steel channels and bonnets shall be postweld heat treated when specified by the purchaser.

6.6.6 Postweld heat treatment is required for all carbon steel floating-head covers that are fabricated by welding a dished-only head into a ring flange.

● **6.6.7** The purchaser shall specify when heat treatment is required for process reasons.

6.7 Dimensional Tolerances

6.7.1 Manufacturing tolerances shall be such that nominally identical parts are interchangeable.

6.7.2 Exchangers that are to be stacked in service shall be stacked in the shop to check connection alignment.

6.7.3 For stacked exchangers, mating nozzle flanges shall not be out of parallel with each other by more than 1/32 inch (0.8 millimeter), measured across any diameter. Separation of mating nozzle flanges shall not exceed 1/8 inch (3 millimeters) after installation of the gasket. Bolts shall be capable of being inserted and removed freely without binding. Shims shall be installed as required between the supports and shall be tack welded in place.

6.8 Gasket Contact Surfaces Other Than Nozzle Flange Facings

6.8.1 Gasket contact surfaces shall have finishes equivalent to those shown in Table 2.

6.8.2 The flatness tolerance (maximum deviation from a plane) on peripheral gasket contact surfaces shall be 1/32 inch (0.8 millimeter). The use of a straightedge to determine flatness tolerance is acceptable.

● **6.8.3** For special applications, such as high-pressure, high-temperature, or hydrogen service, the requirements of 6.8.3.1 and 6.8.3.2 shall be met when specified by the purchaser.

6.8.3.1 The flatness tolerances on peripheral gasket contact surfaces shall be as indicated in Table 3.

6.8.3.2 For exchangers without internal pass partition covers, the flatness tolerance on individual pass partition grooves shall be 1/32 inch (0.8 millimeter).

6.8.4 Flange flatness tolerance and surface finish shall be measured after the flange is attached to the component cylinder or the cover and after any postweld heat treatment.

6.8.5 The flatness of tubesheet gasket contact surfaces shall be measured after the tube-to-tubesheet joints are expanded or welded.

6.9 Tube–Hole Grooves

6.9.1 Tube-hole grooves shall be square edged, concentric, and free from burrs.

6.9.2 For shell-side clad tubesheets, the tube shall be expanded to seal against the cladding material. In no case shall the rolling approach the shell-side face of the tubesheet closer than 1/8 inch (3 millimeters).

Table 2—Gasket Contact Surface Finishes

Type	Roughness
Solid flat metal gaskets	60–125 microinches R_a (AARH)
Double-jacketed gaskets, spiral-wound gaskets	125–250 microinches R_a (AARH)

Note: R_a = roughness average in microinches; AARH = arithmetic average roughness height in microinches.

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Table 3—Flatness Tolerance on Peripheral Gasket Contact Surfaces

Nominal Exchanger Diameter		Tolerance	
Inches	Millimeters	Inches	Millimeters
<15	<375	0.003	0.08
15–30	375–750	0.006	0.15
31–45	751–1125	0.009	0.23
>45	>1125	0.012	0.30

6.9.3 When austenitic stainless steel, duplex stainless steel, titanium, cupro-nickel, or nickel alloy tubes are specified, the tube holes shall be machined in accordance with Table RCB-7.41, column (b) (Special Close Fit), of the TEMA Standards.

6.10 Assembly

6.10.1 Match marks or dowels shall be provided to prevent misassembly of the following bolted joints:

a. Floating-head cover to tubesheet.
b. Channel to tubesheet.
c. Grooved channel cover to channel.
d. Stationary tubesheet to shell.

6.10.2 The threads of external studs and nuts shall be coated with a suitable thread lubricant to prevent galling.

SECTION 7—INSPECTION AND TESTING

7.1 Quality Assurance

7.1.1 Materials, fabrication, conformance with mechanical design, and testing of heat exchangers shall be subject to inspection by the purchaser, his or her designated representative, or both.

7.1.2 The vendor shall notify the purchaser not less than 5 days before the start of fabrication. The vendor shall also notify the purchaser of the date for bundle insertion, other required inspection points, and final inspection.

7.1.3 Unless otherwise specified, the inspector designated by the purchaser shall be permitted entry to the vendor's shop while the work is being performed. Unless otherwise specified, the vendor shall extend to the inspector, the facilities and any reasonable opportunity to ensure that the exchanger is being furnished in accordance with the requirements specified by the purchase order.

7.1.4 No unit shall be released for shipment without the approval of the purchaser or his or her designated representative.

● **7.1.5** When inspection is specified by the purchaser, it shall consist of the following as a minimum:

a. Verifying that qualified welding procedures and qualified welders and welding operators are being used by the manufacturer.
b. Verifying that the construction complies with the applicable drawings and with this standard.
c. Reviewing and/or examining the results of any specified nondestructive testing.
● d. Witnessing hydrostatic testing and any additional testing specified by the purchaser.
e. Examining required certified material test reports and the manufacturer's data report.

7.1.6 No tubes or tube holes shall be plugged without notifying the purchaser. The method and procedure of plugging shall be subject to approval of the purchaser.

7.2 Quality Control

7.2.1 If full radiography is not specified or required by Section VIII, Division 1, of the ASME Code, the exchanger or component shall be spot radiographed as follows:

a. At least one spot radiograph shall be made of each Category A and B joint (as defined by Section VIII, Division 1, of the ASME Code). Nozzle welds are excluded from this requirement.
b. Spot radiographs shall include each start and stop of welds made by the automatic submerged-arc welding process.
c. Spot radiographs shall be a least 10 inches (254 millimeters) long or shall be full length where the weld is less than 10 inches (254 millimeters) long.
d. Spot radiographs shall not exceed porosity standards of Section VIII, Division 1, of the ASME Code for fully radiographed joints.

7.2.2 The magnetic-particle examination and the criteria for acceptance shall comply with Appendix 6 of Section VIII, Division 1, of the ASME Code.

7.2.3 For nonmagnetic materials, a liquid-penetrant examination shall be used in place of any required magnetic-particle examination.

7.2.4 The liquid-penetrant examination and the criteria for acceptance shall comply with Appendix 8 of Section VIII, Division 1, of the ASME Code.

7.2.5 The requirements for weld hardness testing are as follows:

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a. The weld metal and heat-affected zone of pressure retaining welds in components made from a material that has a P number of 1, 3, 4, 5, 6, or 7 shall be tested.
b. Examination shall be made after any required postweld heat treatment.
c. Hardness shall not exceed 225 Brinell for materials with P numbers of 1, 3, or 4; or 240 Brinell for materials with P numbers of 5, 6, or 7.
d. Hardness shall be determined using a 10-millimeter diameter ball unless otherwise specified.
e. One longitudinal weld, one circumferential weld, and each connection-to-component weld where the connection is NPS 2 or larger shall be tested.
f. If more than one welding procedure is used to fabricate longitudinal or circumferential welds, hardness readings shall be made of welds deposited by each procedure.

7.2.6 For alloy clad construction with a P number base metal of 3 and higher, both the stripped section of the base metal and the weld shall be examined by magnetic-particle method prior to applying weld overlay.

7.2.7 All finished welds in material of ferromagnetic steel with a P number of 3 or higher shall be examined after postweld heat treatment (unless the ASME Code specifies examination after hydrostatic testing) by the magnetic-particle method.

7.2.8 Final welds in all nonmagnetic materials, whether of solid alloy or alloy-clad plate, shall be examined by the liquid-penetrant method after any required postweld heat treatment.

7.2.9 Final visual weld inspection shall be performed after postweld heat treatment.

7.3 Pressure Testing

7.3.1 Except for differential pressure designs, an independent hydrostatic test of shell-side and tube-side shall be performed. The minimum fluid temperature for the hydrostatic testing shall be as recommended by Section VIII, Division 1, of the ASME Code.

7.3.2 The hydrostatic test pressure shall be maintained for at least 1 hour. After completion of the test, the exchanger shall be drained.

7.3.3 Water used for hydrostatic testing of units in which austenitic stainless steel materials will be exposed to the test fluid shall be potable water with a chloride ion content of less than 50 parts per million.

● **7.3.4** When specified by the purchaser, the shell-side hydrostatic test shall be conducted with the bonnet or channel cover removed.

7.3.5 Nozzle reinforcement pads shall be pneumatic tested at 25 pounds per square inch (170 kilopascals) gauge.

7.3.6 For safety considerations, any supplementary pneumatic test shall be performed at a nominal pressure of 25 pounds per square inch (170 kilopascals) gauge.

● **7.3.7** Any other test requirements, such as the use of a test ring, shall be mutually agreed upon by the purchaser and the vendor.

7.3.8 Flanged joints that have been taken apart after a hydrostatic test shall be reassembled with unused gaskets and rehydrotested for tightness.

7.3.9 Paint or other external coatings shall not be applied over welds before the final hydrostatic test.

7.3.10 Exchangers that are stacked in service shall be hydrotested as a stacked unit.

7.4 Nameplates and Stampings

7.4.1 A stainless steel nameplate shall be permanently attached to the exchanger in such a manner that it is visible after insulating.

7.4.2 The nameplate shall be located on the shell near the channel end.

7.4.3 The following parts shall be stamped with the manufacturer's serial number:

a. Shell flange.
b. Shell cover flange.
c. Channel or bonnet flange.
d. Channel cover.
e. Stationary tubesheet.
f. Floating tubesheet.
g. Floating-head cover flange.
h. Floating-head backing device.
i. Test ring flange and gland.

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SECTION 8—PREPARATION FOR SHIPMENT

8.1 Protection

8.1.1 All openings in the exchanger shall be suitably protected to prevent damage and the possible entry of water and other foreign material.

8.1.2 All flange gasket surfaces shall be coated with an easily removable rust preventative and shall be protected by suitably attached durable covers such as wood, plastic, or gasketed steel.

8.1.3 All threaded connections shall be protected by metal plugs or caps of compatible material.

8.1.4 Connections that are beveled for welding shall be suitably covered to protect the bevel from damage.

● **8.1.5** All exposed ferrous surfaces not otherwise coated shall be given one coat of manufacturer's standard shop primer. Any additional painting requirements shall be specified by the purchaser.

8.2 Identification

8.2.1 The item number, shipping weight, and purchaser's order number shall be painted on the exchanger.

8.2.2 All boxes, crates, or packages shall be identified with the purchaser's order number and the equipment item number.

8.2.3 Stencil "DO NOT WELD" (in two places 180 degrees apart, as a minimum) on equipment that has been post-weld heat treated.

● SECTION 9—SUPPLEMENTAL REQUIREMENTS

9.1 General

This section discusses additional design, fabrication, and examination requirements that apply to one or both sides of the exchanger when specified by the purchaser. In general, these supplemental requirements should be considered when the cylinder thickness of an exchanger component exceeds 2 inches (50 millimeters) or when an exchanger is to be placed in a critical service.

9.2 Design

9.2.1 Connections to components shall conform to Figure UW 16.1, sketches (a), (b), (c), (d), (e), (f-1), (f-2), (f-3), or (f-4) of Section VIII, Division 1, of the ASME Code. However, connections of the types shown in sketches (a) and (b) shall be made only to forged components or to plates ultrasonically examined in the area of attachment for a radial distance twice the thickness of the component.

9.2.2 Tubesheet attachment welds shall be butt welds.

9.3 Examination

9.3.1 All material for formed heads or cylinders exceeding 2 inches (50 millimeters) in thickness shall be ultrasonically examined. The examination and the criteria for acceptance shall comply with Specification SA-435 and Article 23 of Section V of the ASME Code.

9.3.2 All forgings except standard flanges designed in accordance with ASME B16.5 or ASME B16.47 shall be ultrasonically examined in accordance with Specification SA-388 and Article 23 of Section V of the ASME Code. The criteria for acceptance shall be agreed upon by the purchaser and the vendor.

9.3.3 For ultrasonic examination of welds and forgings, the purchaser shall be supplied with a report providing diagrams of the surfaces scanned and indications obtained, the areas repaired, the nature of defects repaired, and the repair procedures used. The following information also shall be provided:

a. The pulse-echo unit manufacturer, model, and damping control setting.
b. The search unit manufacturer, model, dimensions, and couplant.
c. The frequency used and the test angle to the component surface.
d. The wedge medium for angle-beam examination.

9.3.4 A magnetic-particle examination shall be performed on all plate edges and openings before welding. Any defects found shall be removed and any necessary repairs performed.

9.3.5 A magnetic-particle examination shall be performed on all pressure-retaining welds. If accessible, the back side of the root pass shall be examined after being prepared for final welding. Both sides of accessible completed welds shall be examined.

9.3.6 A magnetic-particle examination shall be performed on all pressure-boundary attachment welds.

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9.3.7 A magnetic-particle examination shall be performed on areas where temporary lugs have been removed. These areas shall be prepared for examination by grinding.

9.3.8 After the hydrostatic test, a magnetic-particle examination shall be performed on all exterior pressure-retaining welds and all interior nozzle welds that are accessible without disassembling the exchanger.

9.3.9 On components subject to full radiography, nozzle-attachment welds that cannot be radiographed shall be examined for the presence of cracks by the magnetic-particle method or by the liquid-penetrant method. Examination shall apply to the root pass after back chipping or after flame gouging where applicable, and to the completed weld. Any defects revealed shall be removed before the weld is finished. Neither the dye nor the developer shall contain any chloride.

9.3.10 A full radiographic examination shall be performed on all pressure-retaining butt welds.

9.3.11 An ultrasonic examination shall be performed on all pressure-retaining butt welds after postweld heat treatment. Ultrasonic examination shall conform with Appendix 12 of Section VIII, Division 1, of the ASME Code. The entire volume of deposited weld metal shall be examined from two directions. Before the welds are examined, the adjacent base material shall be examined by means of longitudinal beam with a 100-percent scan for a distance of twice the plate thickness back from the weld. A diagram shall be prepared indicating all areas larger than ½ inch (13 millimeters) in diameter that show a loss of back reflection of 50 percent or more. The criteria for acceptance shall be agreed upon by the purchaser and the vendor.

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APPENDIX A—NOTES TO PURCHASER AND VENDOR

In conjunction with this standard, the following documents are intended to provide all data necessary for the description and design of shell-and-tube heat exchangers for general refinery services:

a. Appendix B, Shell-and-Tube Heat Exchanger Checklist.
b. Appendix C, Responsibility Specification Sheet.
c. Appendix D, Shell-and-Tube Heat Exchanger Specification Sheet.
d. The purchaser's supplementary process and physical property data for multicomponent, two-phase services.

Preparation of the specification sheets is a mutual responsibility of the purchaser and the vendor (as shown in Appendix C). The purchaser (owner and/or contractor) is responsible for the pertinent data in Appendix B, which defines the purchaser's explicit requirements for those items in this standard that offer alternatives. Specifications shown with common responsibility should be furnished by the originating party.

The vendor is responsible for the information in Appendix D, as detailed in Appendix C. After the exchanger is fabricated, the vendor shall revise the specification sheet as necessary to "as-built" conditions.

It is intended that the Shell-and-Tube Heat Exchanger Specification Sheet be used without change. The specification sheet was developed cooperatively with TEMA to provide a document that would be acceptable throughout the industry.

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APPENDIX B—SHELL-AND-TUBE HEAT EXCHANGER CHECKLIST

This checklist is to be used for listing the purchaser's specific requirements when the standard provides a choice or designates that a decision must be made.

Paragraph	Item	Requirement
1.2.3	Registration with the National Board of Boiler and Pressure Vessel Inspectors? _____	Yes No
3.1.1	Number of copies of outline drawings required for approval: _____	
3.2.5 Item c	Purchaser to specify which additional design calculations are required: _____	
	_____	
	_____	
3.2.5 Item d	Vibration analysis required? _____	Yes No
3.3	Number of copies of reports and records required: _____	
3.3 Item b	Number of copies of "CERTIFIED AS-BUILT" drawings required: _____	
4.1.1	Purchaser must specify both a maximum design temperature and a minimum design metal temperature (MDMT): _____	(Appendix D, line 34)
4.6.5.4	Continuous tube lanes required? _____	Yes No
4.7.2	Connections beveled for welding acceptable in place of flanged? _____	Yes No
4.7.7	Insulation thickness	
	Shell: _____	
	Channel: _____	
4.7.9	Torquing tool: _____	
4.7.10	Chemical cleaning connections required? _____	Yes No
	Size: _____	
4.8.2	Forged flanges other than welding neck acceptable? _____	Yes No
4.11	Is exchanger exposed to hydrogen at partial pressure exceeding 100 psia? _____	Yes No
	Which side of exchanger is in hydrogen service? ❍ Shell side ❍ Tube side	
5.1	ASME SA-193, Grade B7M, bolts required? _____	Yes No
6.1.3	Extra clearance permissible at transverse baffle? _____	Yes No
6.5.6	Welded tube-to-tubesheet joints required? _____	Yes No
6.6.2	Heat treatment requirements and procedures for U-tube bend sections: _____	
	_____	
	_____	
6.6.5	Postweld heat treatment of clad carbon steel channels and bonnets required? _____	Yes No
6.6.7	Postweld heat treatment for process reason? ❍ Shell side ❍ Tube side	
6.8.3	Special flatness tolerance on gasket contact surfaces required? _____	Yes No
7.1.5	Purchaser's inspection required? _____	Yes No
7.1.5 Item d	Additional testing: _____	
	_____	
	_____	
7.3.4	Removal of bonnet or channel cover for shell-side hydrostatic test required? _____	Yes No
7.3.7	Required tests other than standard hydrostatic: _____	
	_____	
	_____	
8.1.5	Painting requirements: _____	
	_____	
	_____	
9	Supplemental requirements apply to: ❍ Shell side ❍ Tube side	
	Details: _____	
	_____	
	_____	

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APPENDIX C—RESPONSIBILITY SPECIFICATION SHEET

| | | |

APPENDIX D—SHELL–AND–TUBE HEAT EXCHANGER SPECIFICATION SHEET

1	Prepared by:										Job No.		
2	Customer										Reference No.		
3	Address										Proposal No.		
4	Plant Location										Date	Rev.	
5	Service of Unit										Item No.		
6	Size	Type			○ (Horiz.) ○ (Vert.)			Connected in		Parallel		Series	
7	Surf./Unit ○ (Gross) ○ (Eff.)		ft²	Shells/Unit				Surf./Shell ○ (Gross) ○ (Eff.)				ft²	
8	**PERFORMANCE OF ONE UNIT**												
9	Fluid Allocation			Shell Side				Tube Side					
10	Fluid Name												
11	Fluid Quantity, Total		lb/h										
12	Vapor (In/Out)		lb/h										
13	Liquid		lb/h										
14	Steam		lb/h										
15	Water		lb/h										
16	Noncondensable		lb/h										
17	Temperature In/Out		°F										
18	Density L/V		lb/ft³	/				/					
19	Viscosity L/V		cP										
20	Molecular Weight, Vapor on line 12												
21	Molecular Weight, Noncondensable on line 16												
22	Specific Heat Capacity L/V		Btu/lb°F	/				/					
23	Thermal Conductivity L/V		Btu/h·ft·°F	/				/					
24	Latent Heat		Btu/lb@°F	@				@					
25	Inlet Pressure		psig										
26	Velocity		ft/s										
27	Pressure Drop, Allow./Calc.		psi	/				/					
28	Fouling Resistance (Min.)		h·ft²·°F/Btu										
29	Heat Exchanged			Btu/h; MTD (Corrected)								°F	
30	Transfer Rate, Service			Clean								Btu/h·ft²·°F	
31	**CONSTRUCTION OF ONE SHELL**										Sketch (Bundle/Nozzle Orientation)		
32				Shell Side			Tube Side						
33	Design Pressure		psig										
34	Design Temperature/MDMT		°F	/			/						
35	No. Passes per Shell												
36	Corrosion Allowance		in.										
37	Connections	In	NPS										
38	Size and	Out	NPS										
39	Rating	Intermediate	NPS										
40	Tube No.	OD	in.	Thk ○ (Min.) ○ (Avg.)		in.	Length	ft	Pitch	in.	▷ 30	△ 60	□ 90	◇ 45
41	Tube Type			Material			○ (Wld) ○ (Smls)							
42	Shell Material			ID		in.	Shell Cover Material				○ (Integ.) ○ (Remov.)			
43	Channel or Bonnet Material						Channel Cover Material							
44	Stationary Tubesheet Material						Floating Tubesheet Material							
45	Floating-Head Cover Material						Impingement Protection ○ (Yes) ○ (No)							
46	Cross Baffles Material	Type		%Cut ○ (Diam.) ○ (Area)			Spacing: c/c		Inlet		in.			
47	Long Baffles Material			Seal Type										
48	Tube Supports Material			U–Bend			Type							
49	Bypass Seal ○ (Yes) ○ (No)			Tube–Tubesheet Joint ○ (Expd) ○ (StrWld) ○ (Seal Wld)										
50	Expansion Joint ○ (Yes) ○ (No)	Material		Type			Avg. Metal Temp. Shell		°F	Tubes		°F		
51	pv²: Inlet Nozzle			Bundle Entrance			Bundle Exit							
52	Gaskets: Shell Side			Tube Side			Floating Head							
53	Code: ASME		Stamp ○ (Yes) ○ (No)				Nat'l Board Reg. ○ (Yes) ○ (No)		TEMA Class					
54	Weights: Dry			Filled with Water			Bundle					lb		
55	Remarks													
56														
57														
58														
59														
60														

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